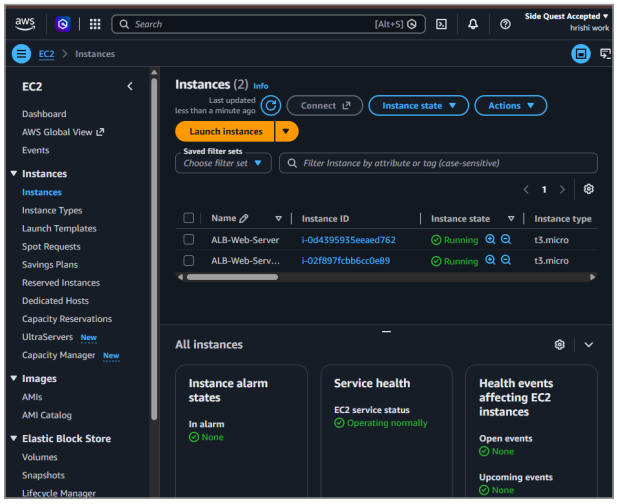
AWS Application Load Balancer (ALB) Practice Tasks

Submitted by Hrishinandan

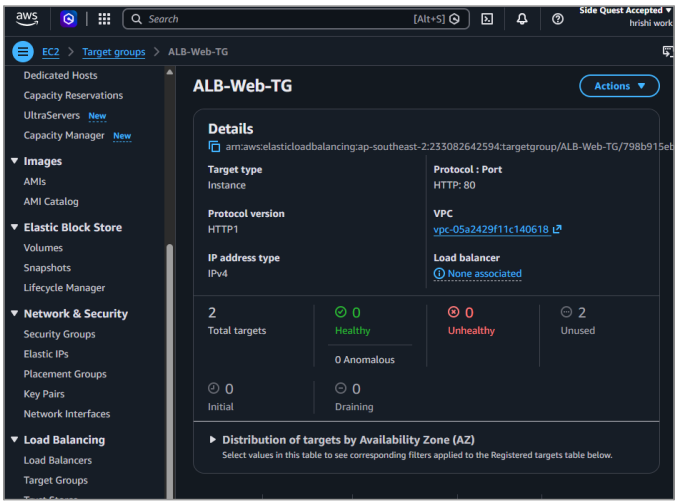
Task 1: Basic Load Balancer

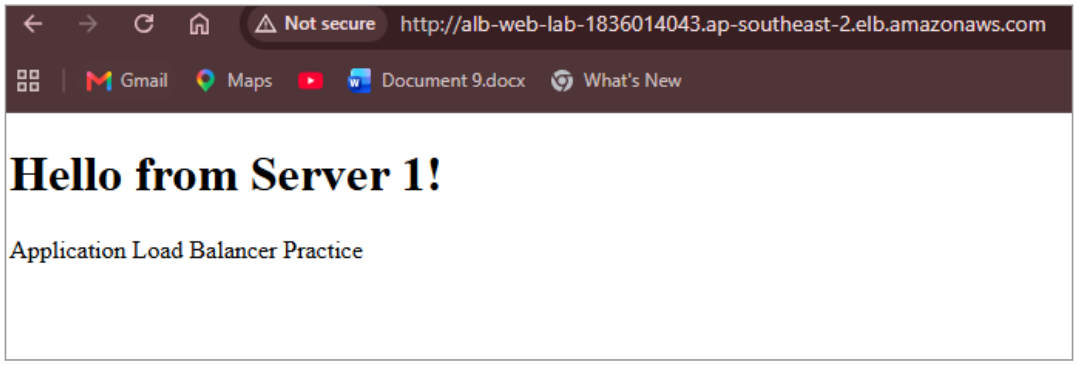
- Launched two EC2 instances with Apache and different home pages, created target group ALB-Web-TG and registered both.
- Created the ALB and attached the target group; refreshing the ALB DNS alternated between Server 1 and Server 2.




```
[ec2-user@ip-172-31-37-75 ~]$ history
    1  sudo systemctl  status httpd
    2  sudo dnf install -y httpd
    3  sudo systemctl enable --now httpd
    4  sudo tee /var/www/html/index.html <<'EOF'
<!DOCTYPE html>
<html>
<head><title>ALB Lab</title></head>
<body>
<h1>Hello from Server 1!</h1>
<p>Application Load Balancer Practice</p>
</body>
</html>
EOF

    5  sudo chmod 755 /var/www/html
    6  sudo chmod 644 /var/www/html/index.html
    7  curl -I http://localhost/
    8  sudo tee /var/www/html/index.html <<'EOF'
<!DOCTYPE html>
<html>
<head><title>ALB Lab</title></head>
<body>
<h1>Hello from Server 1!</h1>
<p>Application Load Balancer Practice</p>
</body>
</html>
EOF
```

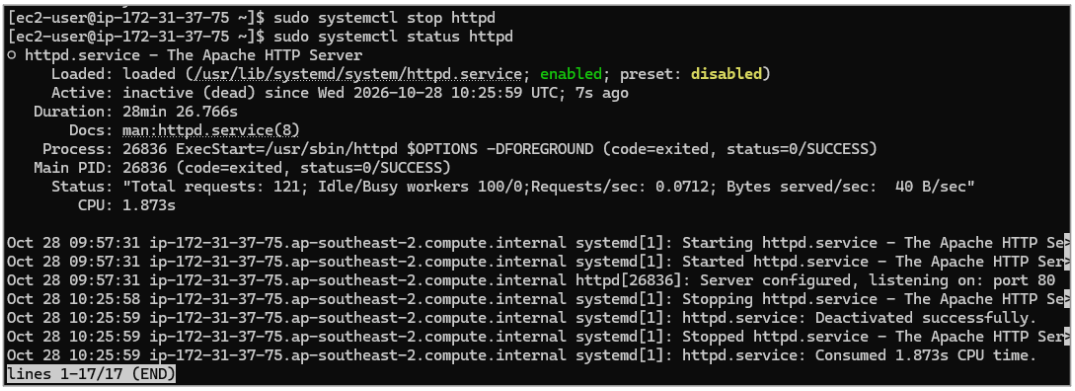

Search
[Alt+S]
Side Quest Accepted
hrishi work
EC2 > Target groups > ALB-Web-TG
Dedicated Hosts
Capacity Reservations
UltraServers New
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Network & Security
Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces
Load Balancing
Load Balancers
Target Groups
ALB-Web-TG
Actions
Details
arn:aws:elasticloadbalancing:ap-southeast-2:233082642594:targetgroup/ALB-Web-TG/798b915eb
Target type
Instance
Protocol : Port
HTTP: 80
Protocol version
HTTP1
VPC
vpc-05a2429f11c140618
IP address type
IPv4
Load balancer
None associated
2
Total targets
0
Healthy
0 Anomalous
0
Unhealthy
2
Unused
0
Initial
0
Draining
Distribution of targets by Availability Zone (AZ)
Select values in this table to see corresponding filters applied to the Registered targets table below.

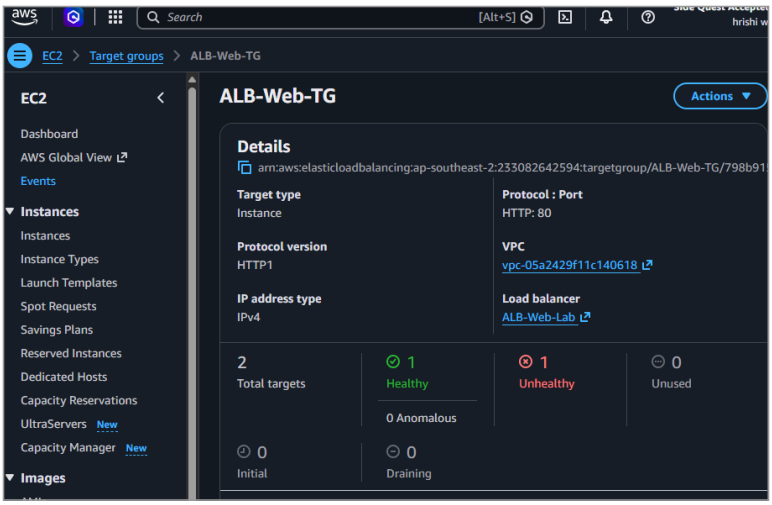

Not secure
http://alb-web-lab-1836014043.ap-southeast-2.elb.amazonaws.com
Gmail
Maps
Document 9.docx
What's New
Hello from Server 1!
Application Load Balancer Practice

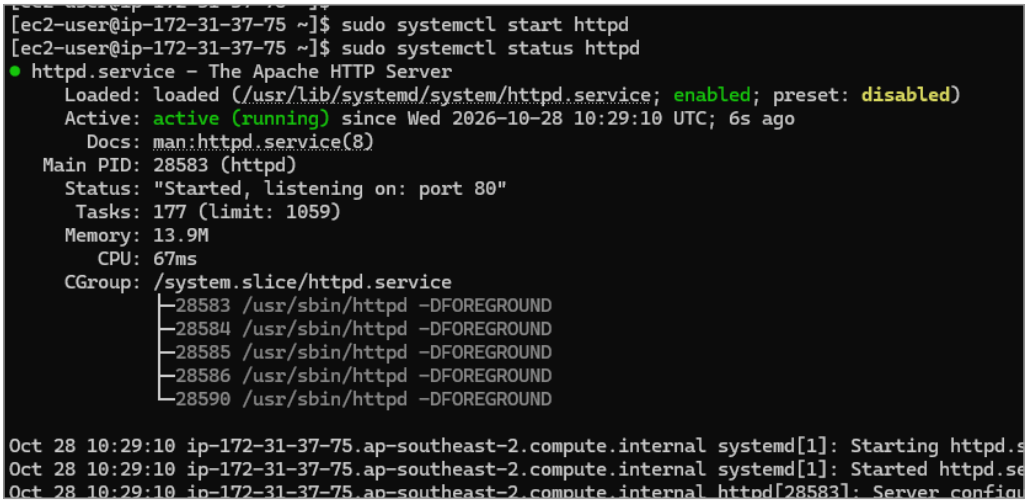

Not secure
http://alb-web-lab-1836014043.ap-southeast-2.elb.amazonaws.com
Gmail
Maps
Document 9.docx
What's New
Hello from Server 2!
Application Load Balancer Practice

Task 2: Health Checks

- Stopped Apache on one instance; the ALB marked it unhealthy.
- Restarted the service and both targets became healthy again.

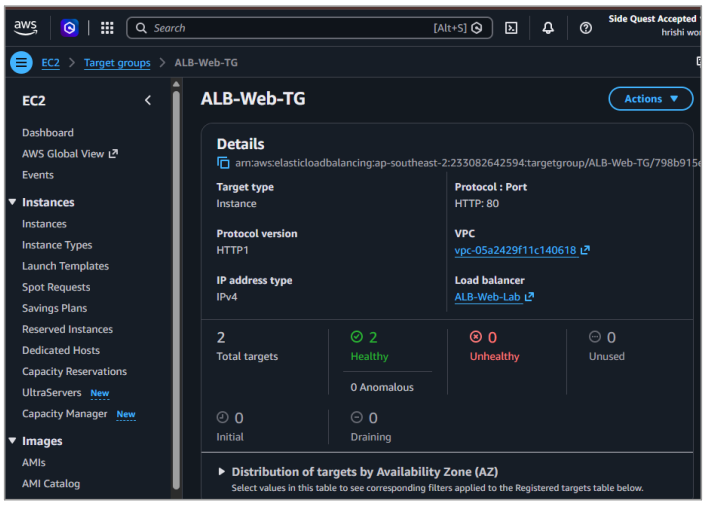






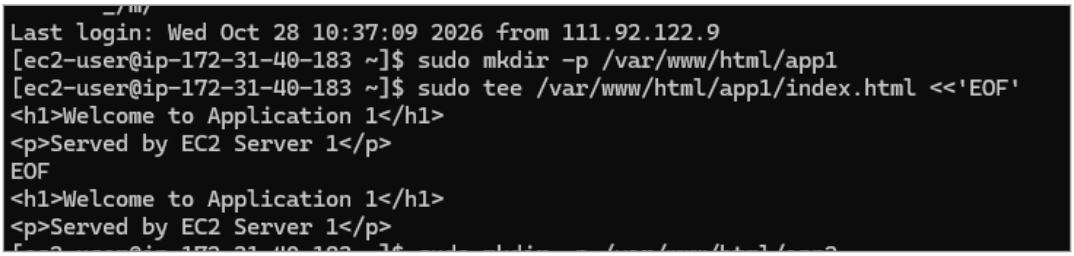
```
[ec2-user@ip-172-31-37-75 ~]$ curl -I http://localhost/
HTTP/1.1 200 OK
Date: Wed, 28 Oct 2026 10:29:38 GMT
Server: Apache/2.4.69 (Amazon Linux)
Last-Modified: Wed, 28 Oct 2026 09:59:26 GMT
ETag: "9a-65ee39e4d24b2"
Accept-Ranges: bytes
Content-Length: 154
Content-Type: text/html; charset=UTF-8

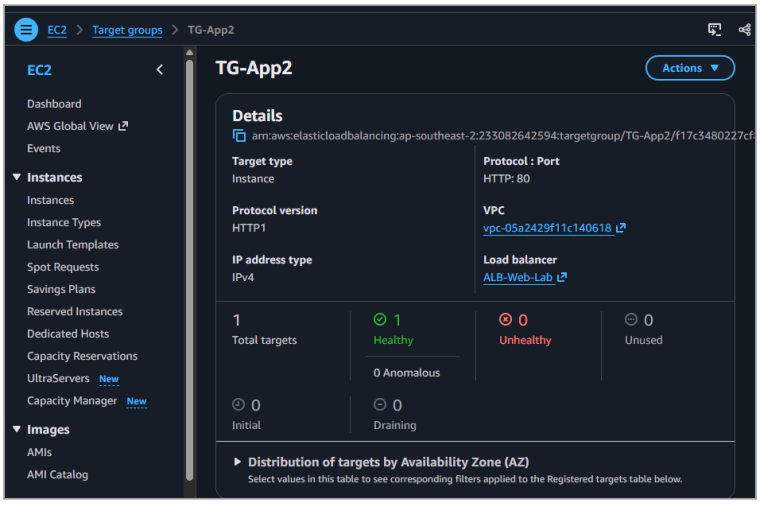
[ec2-user@ip-172-31-37-75 ~]$
```



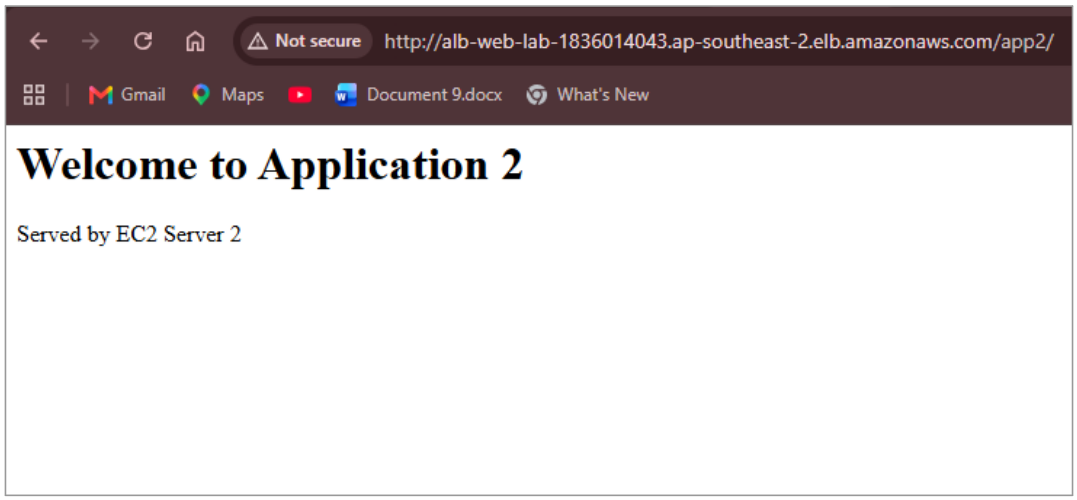
Task 3: Path-Based Routing

- Created two target groups with listener rules: /app1 to one target group and /app2 to the other.
- Browser requests to /app1/ and /app2/ reached Application 1 and Application 2.



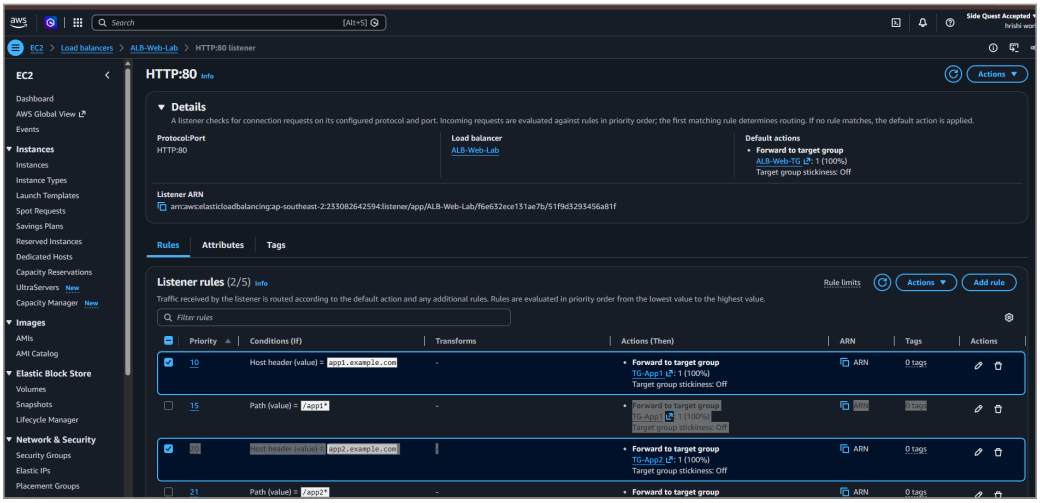





Not secure
http://alb-web-lab-1836014043.ap-southeast-2.elb.amazonaws.com/app2/
Gmail
Maps
Document 9.docx
What's New
Welcome to Application 2
Served by EC2 Server 2

Task 4: Host-Based Routing

- Added host-header listener rules: app1.example.com and app2.example.com each route to their own target group.
- Requests with each Host header were served by Server 1 and Server 2 respectively.




```
C:\Users\hrish>curl -i -H "Host: app1.example.com" http://ALB-Web-Lab-1836014043.ap-southeast-2.elb.amazonaws.com/
HTTP/1.1 200 OK
Date: Thu, 29 Oct 2026 05:41:09 GMT
Content-Type: text/html; charset=UTF-8
Content-Length: 154
Connection: keep-alive
Server: Apache/2.4.69 (Amazon Linux)
Last-Modified: Wed, 28 Oct 2026 09:54:03 GMT
ETag: "9a-65ee38b071dbc"
Accept-Ranges: bytes

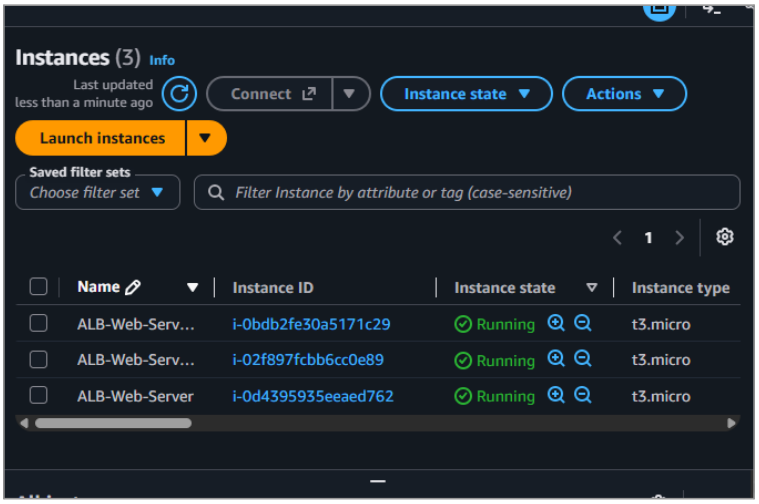
<!DOCTYPE html>
<html>
<head><title>ALB Lab</title></head>
<body>
<h1>Hello from Server 1!</h1>
<p>Application Load Balancer Practice</p>
</body>
</html>
```

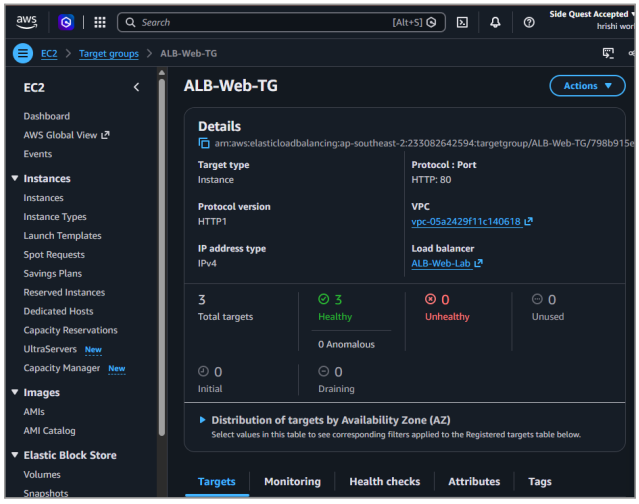
```
C:\Users\hrish>curl -i -H "Host: app2.example.com" http://ALB-Web-Lab-1836014043.ap-southeast-2.elb.amazonaws.com/
HTTP/1.1 200 OK
Date: Thu, 29 Oct 2026 05:41:29 GMT
Content-Type: text/html; charset=UTF-8
Content-Length: 154
Connection: keep-alive
Server: Apache/2.4.69 (Amazon Linux)
Last-Modified: Wed, 28 Oct 2026 09:59:26 GMT
ETag: "9a-65ee39e4d24b2"
Accept-Ranges: bytes

<!DOCTYPE html>
<html>
<head><title>ALB Lab</title></head>
<body>
<h1>Hello from Server 2!</h1>
<p>Application Load Balancer Practice</p>
</body>
</html>
```

Task 5: Add New Server

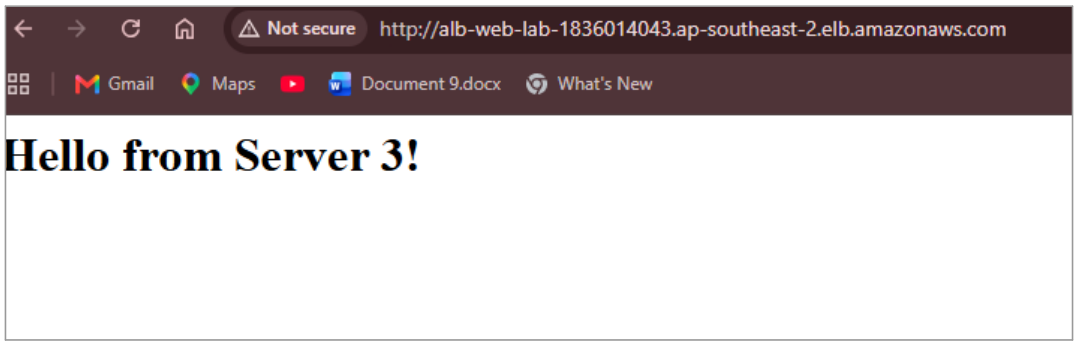
- Launched a third EC2 instance and registered it in the target group; all 3 targets are healthy.
- The ALB DNS served Server 1, Server 2 and Server 3.





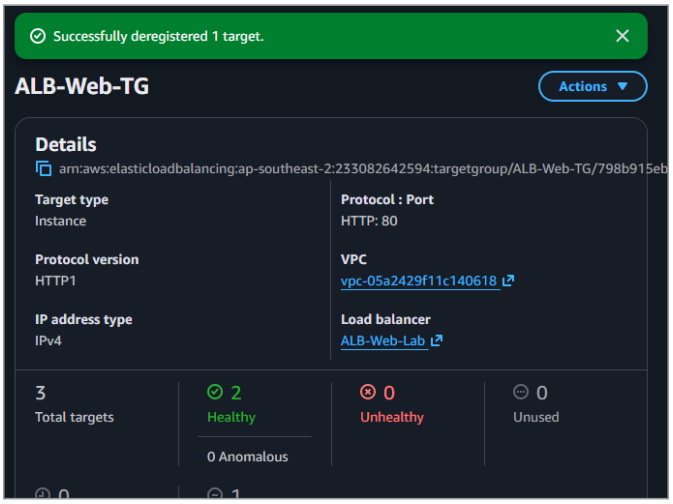


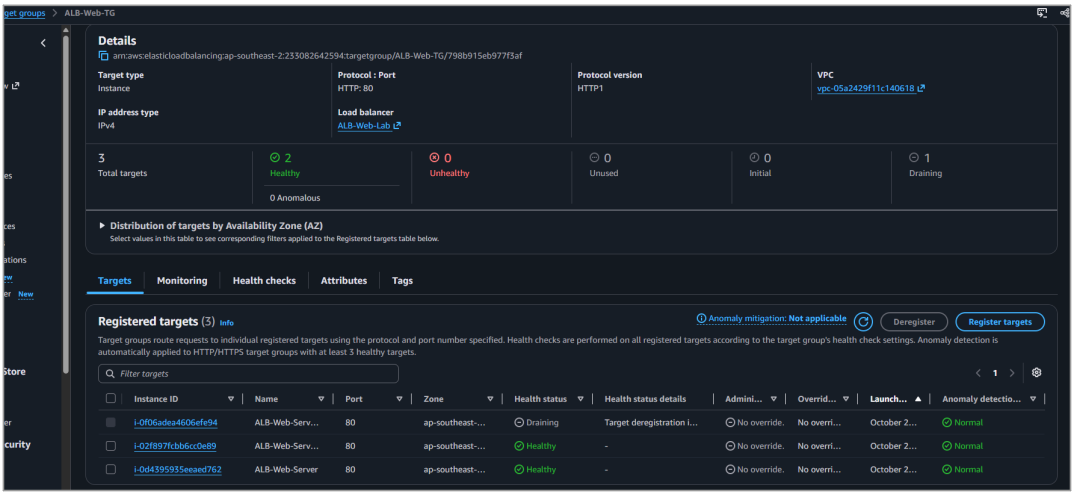

Not secure
http://alb-web-lab-1836014043.ap-southeast-2.elb.amazonaws.com
Gmail
Maps
Document 9.docx
What's New
Hello from Server 2!
Application Load Balancer Practice


Not secure
http://alb-web-lab-1836014043.ap-southeast-2.elb.amazonaws.com
Gmail
Maps
Document 9.docx
What's New
Hello from Server 3!

Task 6: Deregister Instance

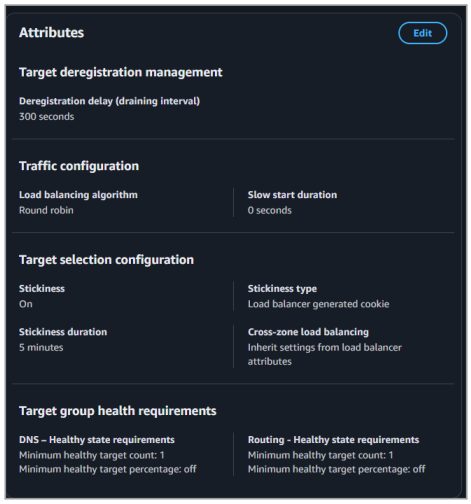
- Deregistered one instance from the target group.
- It moved to Draining while the other two instances stay healthy.





Task 7: Sticky Sessions

- Enabled stickiness on the target group (load balancer generated cookie, 5 minutes).
- Repeated requests with the same cookie always reached Server 2.




```
[ec2-user@ip-172-31-41-79 ~]$ curl -i -c cookies.txt -b cookies.txt http://ALB-Web-Lab-1836014043.ap-southeast-2.elb.amazonaws.com/
HTTP/1.1 200 OK
Date: Thu, 29 Oct 2026 06:51:41 GMT
Content-Type: text/html; charset=UTF-8
Content-Length: 154
Connection: keep-alive
Set-Cookie: AWSALB=I9xmvAy9PDxVnP4fRz8wNukcZwZZlqmgbshpnC3FYO4s3CsoYToAvyKmcW+cUnYnj8NDBpOD1/IdQW9Nqkwq9S3C3Mm4squWHk4xarUOhnCZTnv+b7Oq2Ieb2UWl; Expires=Th
Set-Cookie: AWSALBCORS=I9xmvAy9PDxVnP4fRz8wNukcZwZZlqmgbshpnC3FYO4s3CsoYToAvyKmcW+cUnYnj8NDBpOD1/IdQW9Nqkwq9S3C3Mm4squWHk4xarUOhnCZTnv+b7Oq2Ieb2UWl; Expire
Server: Apache/2.4.69 (Amazon Linux)
Last-Modified: Wed, 28 Oct 2026 09:59:26 GMT
ETag: "9a-65ee39e4d24b2"
Accept-Ranges: bytes

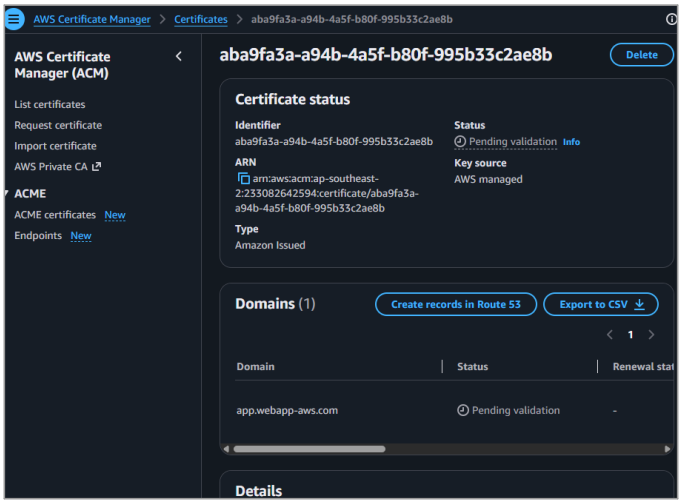
<!DOCTYPE html>
<html>
<head><title>ALB Lab</title></head>
<body>
<h1>Hello from Server 2!</h1>
<p>Application Load Balancer Practice</p>
</body>
</html>
```

```
[ec2-user@ip-172-31-41-79 ~]$ curl -i -c cookies.txt -b cookies.txt http://ALB-Web-Lab-183601404
HTTP/1.1 200 OK
Date: Thu, 29 Oct 2026 06:54:06 GMT
Content-Type: text/html; charset=UTF-8
Content-Length: 154
Connection: keep-alive
Set-Cookie: AWSALB=w4ejgZiU3OpRFTTKHQwBHQ43ShUS96FuiJb3A9wXoFCU99Paac62sWkgWoOQkqN+yFCmVnjEWFQRF
Set-Cookie: AWSALBCORS=w4ejgZiU3OpRFTTKHQwBHQ43ShUS96FuiJb3A9wXoFCU99Paac62sWkgWoOQkqN+yFCmVnjEW
Server: Apache/2.4.69 (Amazon Linux)
Last-Modified: Wed, 28 Oct 2026 09:59:26 GMT
ETag: "9a-65ee39e4d24b2"
Accept-Ranges: bytes

<!DOCTYPE html>
<html>
<head><title>ALB Lab</title></head>
<body>
<h1>Hello from Server 2!</h1>
<p>Application Load Balancer Practice</p>
</body>
</html>
[ec2-user@ip-172-31-41-79 ~]$
```

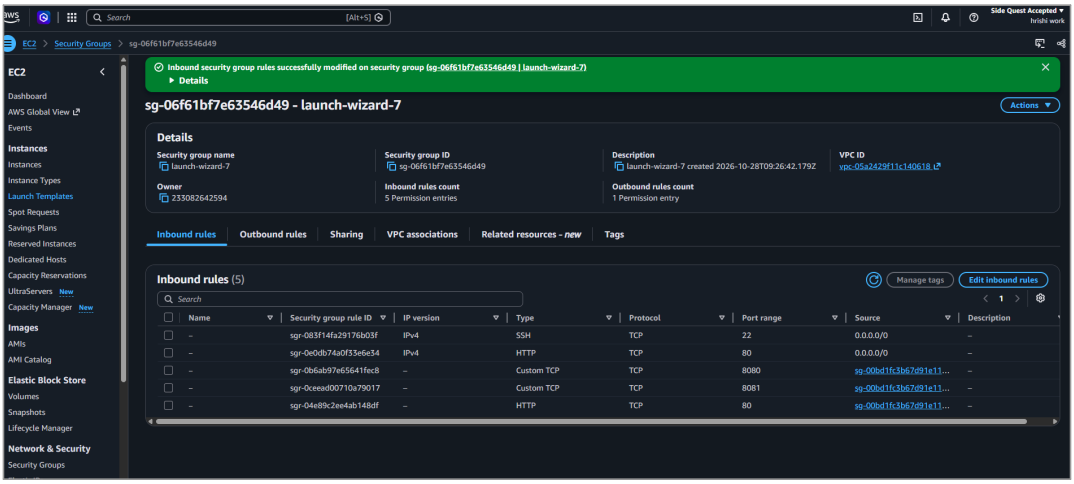
Task 8: HTTPS

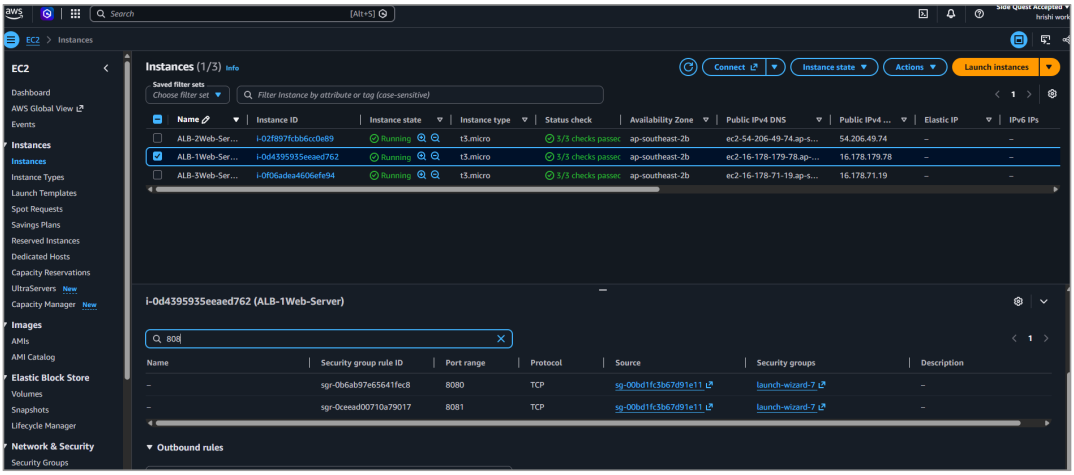
- Requested an ACM certificate for app.webapp-aws.com.
- Status is Pending validation, as no registered domain was available.



Task 9: Multi-Port Routing

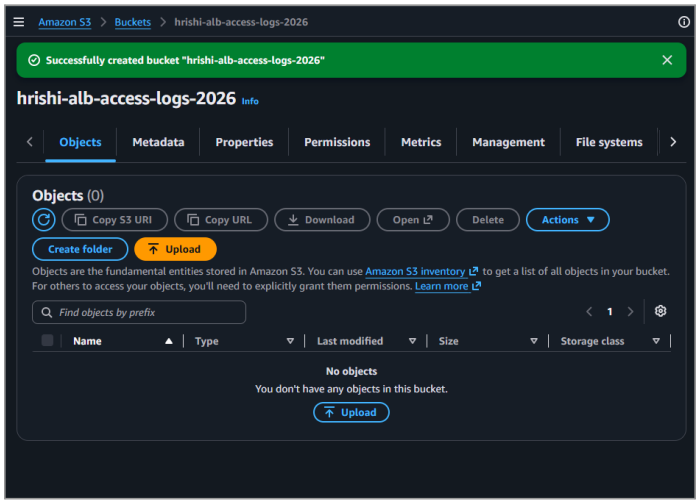
- Opened ports 8080 and 8081 in the instance security group with the ALB security group as source.
- Both rules are confirmed on the instance's security tab.

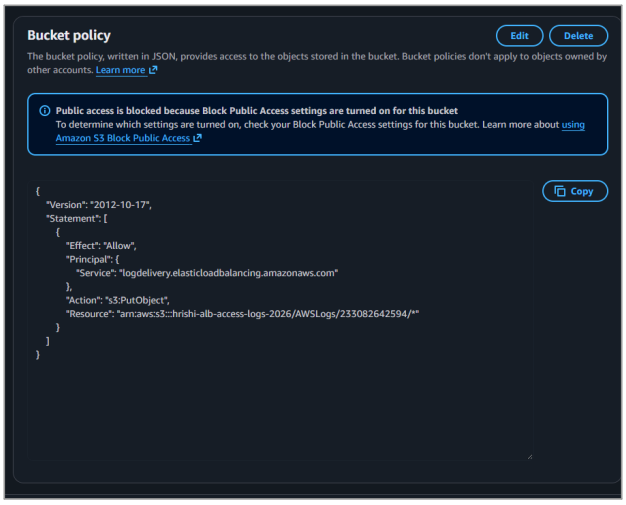


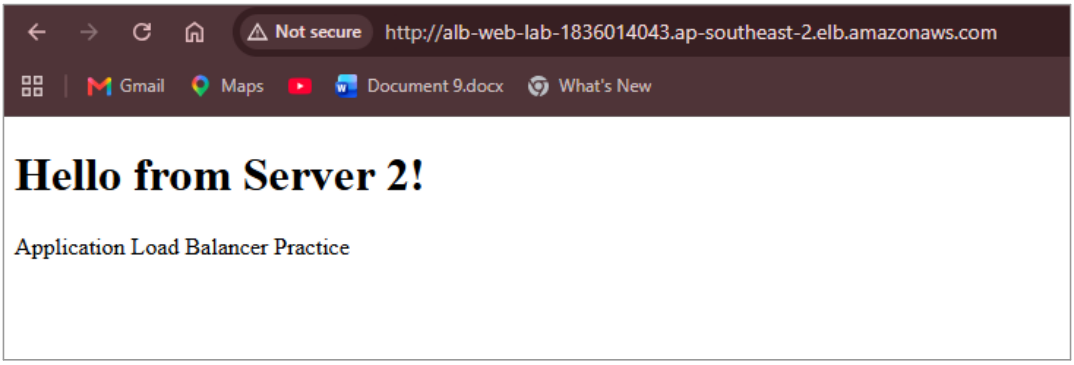


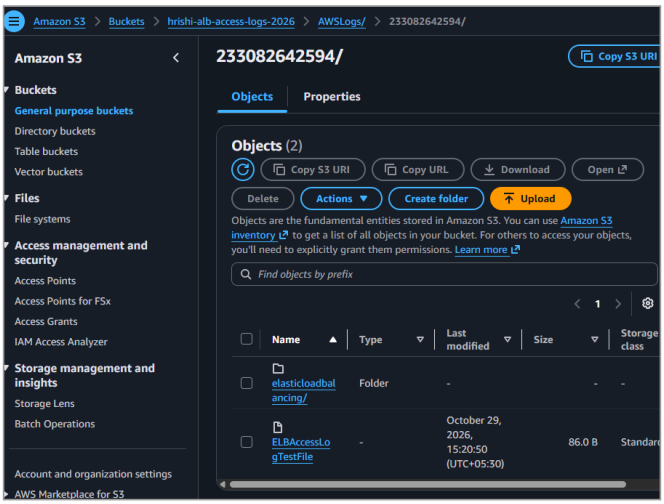
Task 10: Access Logs

- Created an S3 bucket with a bucket policy that allows ELB log delivery.
- Generated traffic through the ALB DNS; logs were delivered under AWSLogs in the bucket.



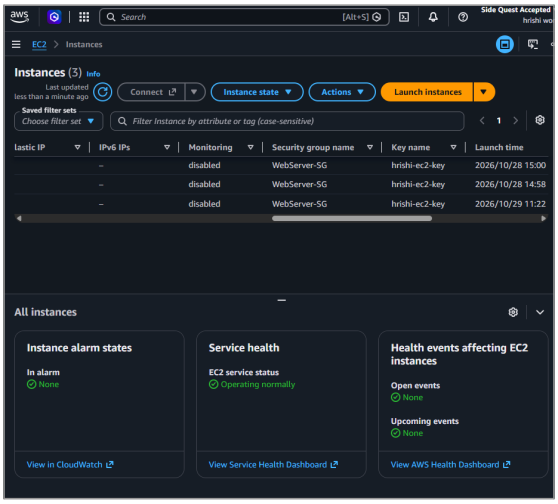


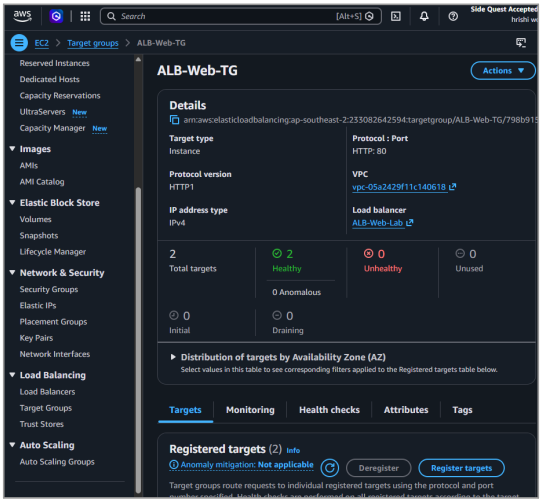



Amazon S3 > Buckets > hrishi-alb-access-logs-2026 > AWSLogs/ > 233082642594/
Amazon S3
Buckets
General purpose buckets
Directory buckets
Table buckets
Vector buckets
Files
File systems
Access management and security
Access Points
Access Points for FSx
Access Grants
IAM Access Analyzer
Storage management and insights
Storage Lens
Batch Operations
Account and organization settings
AWS Marketplace for S3
233082642594/
Copy S3 URI
Objects
Properties
Objects (2)
Copy S3 URI
Copy URL
Download
Open
Delete
Actions
Create folder
Upload
Objects are the fundamental entities stored in Amazon S3. You can use Amazon S3 inventory to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. Learn more
Find objects by prefix
1
Name
Type
Last modified
Size
Storage class
elasticloadbalancing/
Folder
-
-
-
ELBAccessLogTestFile
-
October 29, 2026, 15:20:50 (UTC+05:30)
86.0 B
Standard

Task 11: Security Groups

- Attached the WebServer-SG security group to the EC2 instances.
- Targets remain healthy through the ALB.





Task 12: Auto Scaling

- Created launch template ALB-Web-LT and Auto Scaling Group ALB-Web-ASG (1 to 4 instances).
- The target group shows 3 healthy targets.



